# DIC25
# Nonideal Effects in Dynamic Logic

Specification-based laboratory report

17 October 2026

**English edition**
HSPICE S-2021.09 | Model library supplied separately

## 1 Objectives

Complete the nine dynamic-logic experiments on handout PDF pages 99–106: charge sharing, explicit and parasitic coupling, leakage, three remedies, and power comparison at 500 MHz.

## 2 Handout requirements and test mapping

| Requirement | Deck prefix and conditions | Status |
|---|---|---|
| Step 1: NAND charge sharing | 01; A rises, B=0 | Measured |
| Steps 2–4: CC sweep, inversion, move to output | 02; 5–50 fF | Transition bracket observed |
| Steps 5–6: upper/lower NAND gate coupling | 03; intrinsic coupling | Measured |
| Step 7: leakage during long evaluation | 04; 20 us, 25/115 C | Measured |
| Step 8: always-on/feedback keeper | 04/05/06; 500 MHz | Retention, sharing, power |
| Step 9: internal-node precharge | 04/05 | Sharing/leakage separated |

## 3 Circuit and derivation

The dynamic NAND precharges at low clock and evaluates at high clock. An explicit 5 fF is placed at both output and internal stack nodes for reproducibility, in addition to MOS parasitics. A lossless sharing estimate is $V_f = (C_Y V_{Y0} + C_X V_{X0})/(C_Y + C_X)$, but threshold, body effect, initial levels, and feedthrough prevent exact agreement with an ideal equal-capacitor result.

A rapid aggressor transition approximately induces $\Delta V_Y \simeq C_C \Delta V_A/(C_Y + C_C)$. A driven inverter output has a restoring low-impedance path, whereas a floating dynamic node can retain erroneous charge. Keepers use weak PMOS devices with $W/L = 0.3/2\,\mu$m; the internal precharge PMOS uses $0.6/0.25\,\mu$m.

# 4 Simulation setup and interpretation

The supplied encrypted MOS model is evaluated at a baseline supply of 1.0 V and 25 °C. The 1.0 V value is a teaching condition, not a verified voltage rating. These results do not reproduce the handout's original process or supply conditions. Unless stated in the deck, $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m, bulks are tied to their corresponding rails, and output load is 5 fF. Transmission-gate and weak-feedback dimensions are specified explicitly in each deck.

PrimeSim HSPICE S-2021.09 win64 was used with `RUNLVL=6`, `ACCURATE`, and `WARNLIMIT=100`. Every baseline and `.alter` analysis concluded without errors or failed measurements. The simulator still reports suppressed warnings associated with encrypted blocks; this does not establish that the model is warning-free or that those warnings are harmless. The `.tran` print interval is not a fixed integration step: the solver uses adaptive timesteps. A finite sweep is not a complete process qualification.

Unless stated otherwise, the declared functional diagnostic uses $V \leq 0.2$ V for low and $V \geq 0.8$ V for high. These are report-level interpretation thresholds, not numerical handout requirements or extracted receiver noise margins. Correct samples do not exclude glitches between samples. Figures are redrawn from actual simulation data; `results/` contains per-case waveforms, measurements, and derived comparison CSV files.

Public decks omit model loading paths; model configuration is added only to local execution copies. The model itself is not distributed. The first two filename digits identify the deck numbers used below. Every deck is standalone.

# 5 Results

In the basic sharing case without an output inverter, Y falls from 1.172805 to 0.701010 V, a 0.471795 V drop, while internal X reaches 0.622395 V. The initial value above 1 V is clock feedthrough, not a higher supply. Remedy comparisons use a separate set with matched inverter loading; their baseline droop differs and must not be mixed with this unloaded result.

| Coupling (fF) | Dynamic Y (V) | Inverter output (V) | Direct-output minimum (V) |
|---|---|---|---|
| 5 | 0.8277 | 0.0000 | -0.2632 |
| 10 | 0.6555 | 0.0026 | -0.4427 |
| 15 | 0.5511 | 0.0221 | -0.5547 |
| 20 | 0.4845 | 0.2850 | -0.6161 |
| 30 | 0.4027 | 0.9760 | -0.6562 |
| 50 | 0.2859 | 0.9985 | -0.6843 |

At CC=20 fF, the following inverter output is 0.284995 V, inside this report's conservative indeterminate region. At 30 fF it is 0.976020 V, a clear erroneous high. Using 0.5 V as a receiver decision level brackets the transition between 20 and 30 fF. The same 30 fF coupled directly to the inverter output briefly drives it to -0.656 V, but it recovers to approximately 0 V by 6 ns. Recovery within this window does not mean the spike is harmless to every receiver.

| NAND gate position | Before (V) | After (V) | Drop (V) |
|---|---|---|---|
| upper | 1.13912 | 0.94524 | 0.19388 |
| lower | 1.12564 | 1.07639 | 0.04925 |

With no added CC and only device parasitics, coupling at the upper NMOS gate produces a 0.193876 V Y drop, versus 0.049251 V at the lower gate. This supports the role of gate/source/drain

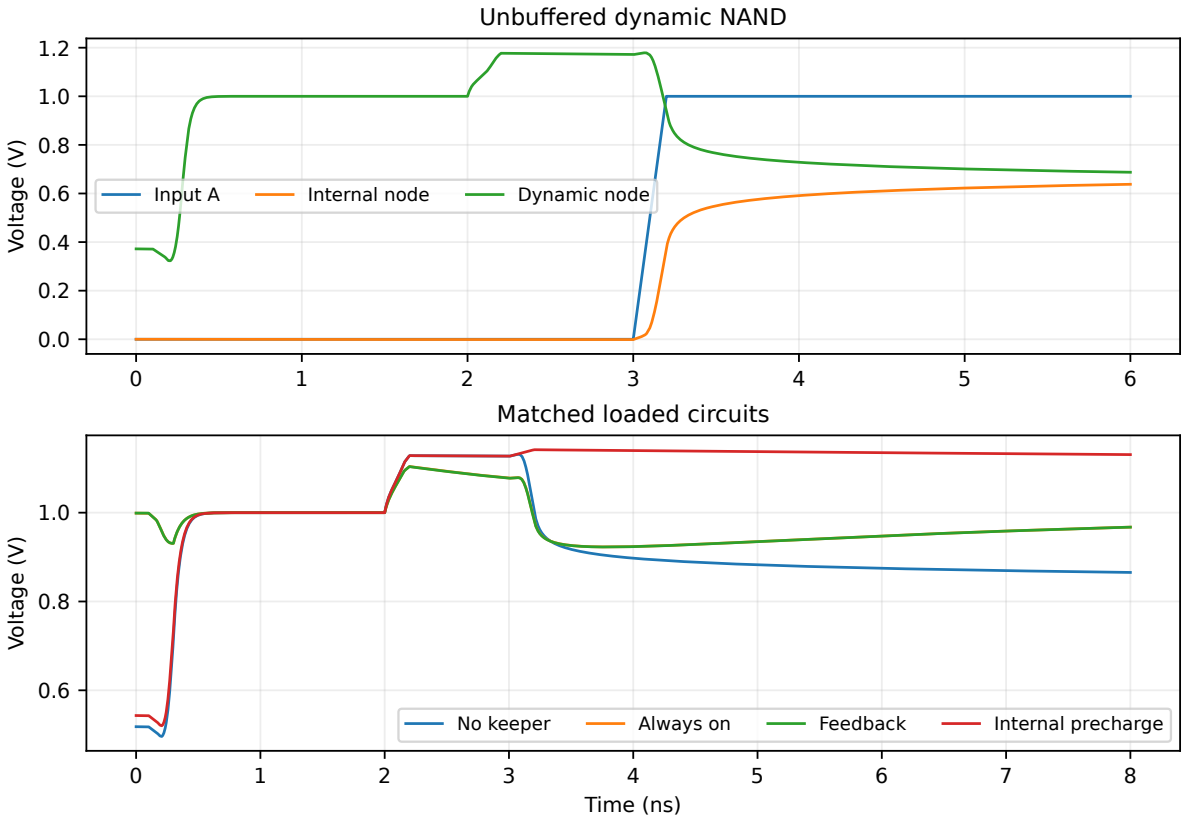



Figure 1: Basic sharing and the three remedies under matched loading.

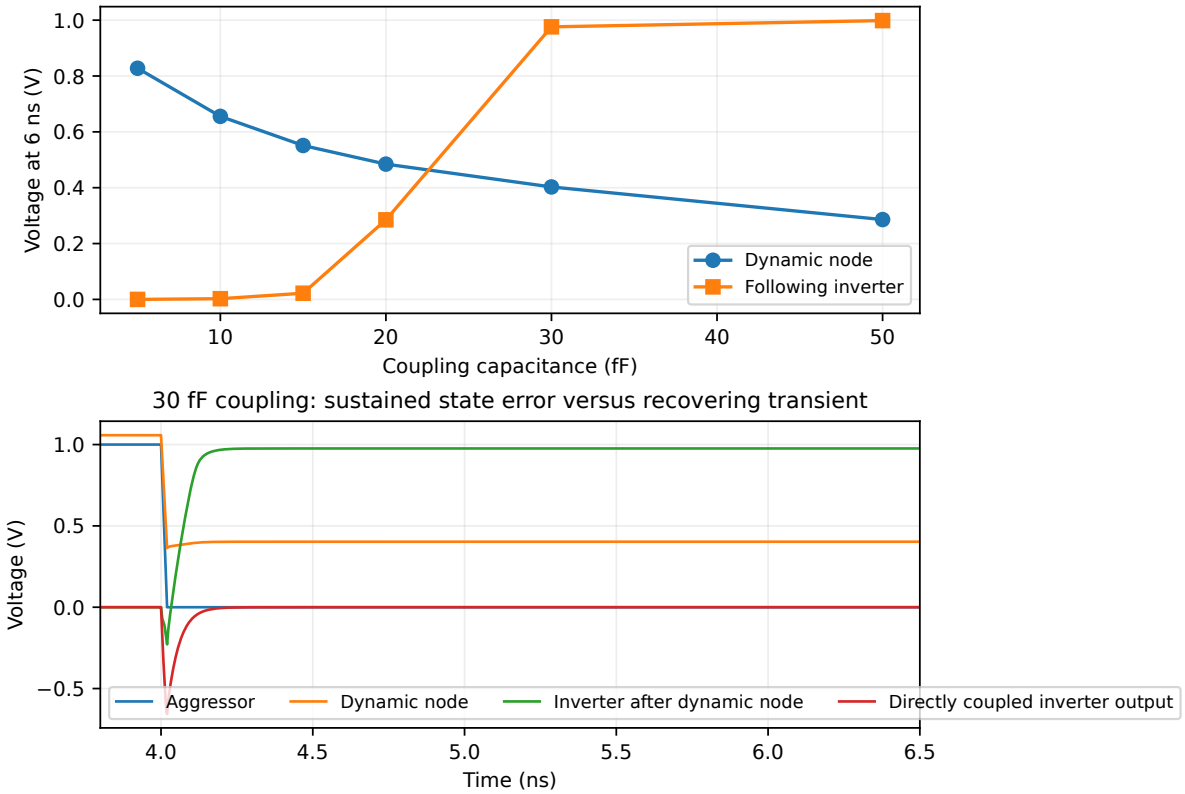



Figure 2: CC sweep and representative 30 fF waveforms, distinguishing sustained error from a recovering spike.

location and intermediate-node shielding: logically equivalent input pins need not have equal analog sensitivity.

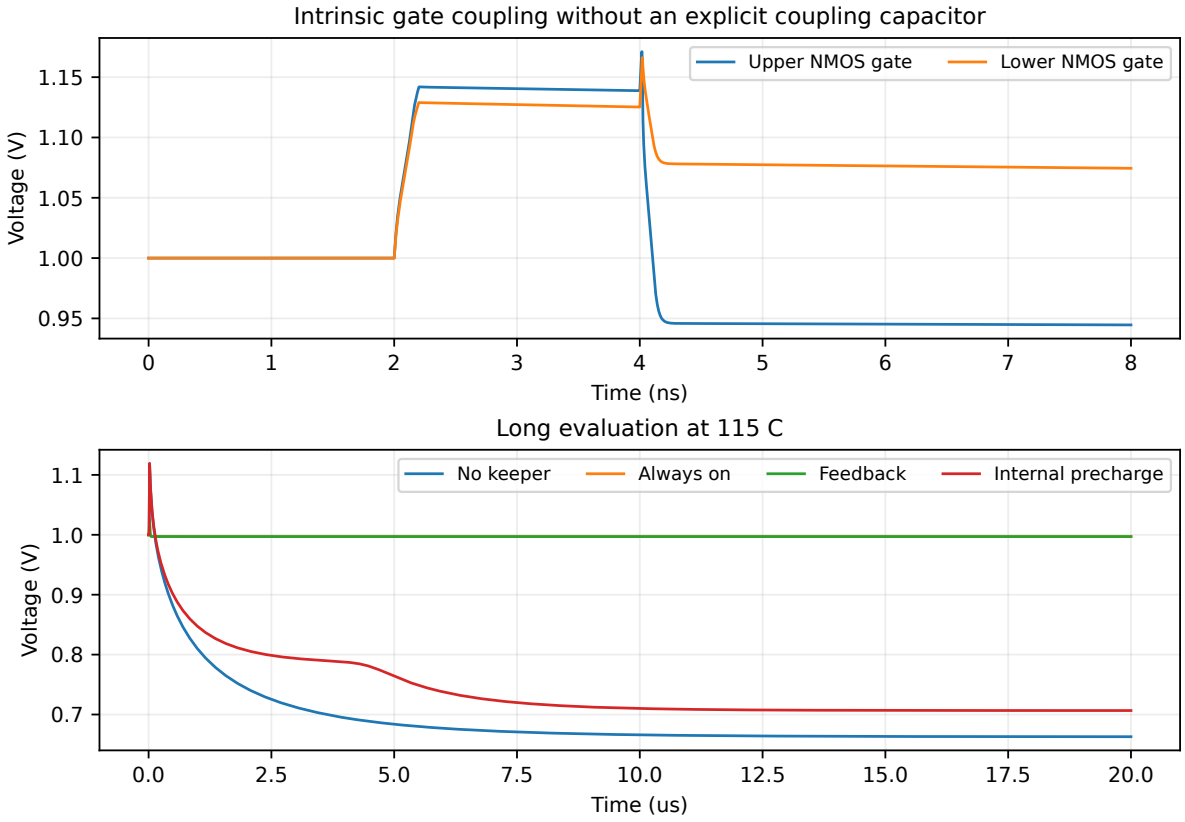



Figure 3: Upper: intrinsic parasitic coupling. Lower: 20 us retention at 115 C.

| Remedy | Sharing drop (V) | Final at 25 C (V) | Final at 115 C (V) |
|---|---|---|---|
| None | 0.24463 | 0.53783 | 0.66294 |
| Always on | 0.14851 | 0.99896 | 0.99702 |
| Feedback | 0.14830 | 0.99896 | 0.99702 |
| Internal precharge | -0.01016 | 0.58577 | 0.70663 |

Always-on and feedback keepers maintain long-hold levels near 0.997–0.999 V. Internal precharge suppresses sharing but does not continuously replenish charge during evaluation. The uncorrected final level is higher at 115 C than at 25 C, showing that net current depends on multiple paths in this model and bias condition. It must not be rewritten as necessarily faster decay at higher temperature. A complete inverter logic flip was not observed within 20 us in this leakage test.

| Freq. (MHz) | Always power (uW) | Feedback power (uW) | Always eval. (uA) | Feedback eval. (uA) |
|---|---|---|---|---|
| 500 | 16.1263 | 16.5449 | 9.9347 | 8.2381 |
| 50 | 2.4854 | 1.6582 | 1.8482 | 0.6664 |

## 6 Discussion, differences, and unverified scope

At 500 MHz, feedback average power is 16.5449 $\mu$W, slightly above 16.1263 $\mu$W for always-on. Feedback must not be declared universally lower-power. Its evaluation current is lower, but cycle-average power also includes dynamic charging, delay, and switching contention. At 50 MHz, feedback

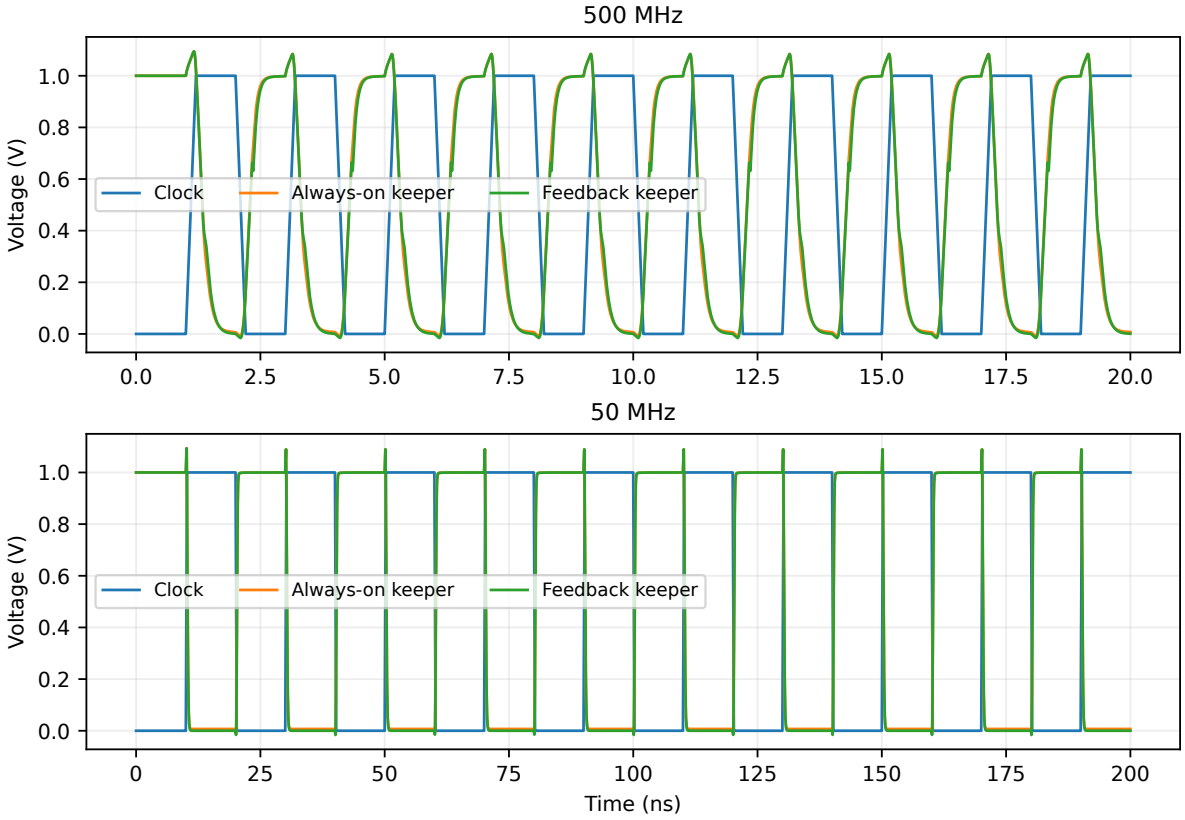



Figure 4: The required 500 MHz test is retained; 50 MHz is added as a more fully settled comparison.

average power falls to 1.6582 $\mu$W versus 2.4854 $\mu$W for always-on. Power is averaged over $2T$–$10T$ and evaluation current over $8.7T$–$8.9T$. All nine observations have corresponding tests; qualitative handout explanations are not converted into invented numerical pass/fail requirements.